

Final Presentations: Guidelines

Logistics

The presentation days are **Tuesday, April 22** and **Thursday, April 24**, from 10:00 through 17:00, in IA-4180. Each pair of students must attend a two-hour time slot, which includes their own presentation. In addition, everyone is welcome to attend both full days if they want. By **Tuesday, April 8** you must have:

1. Signed up for a time slot at the following link:
<https://www.signupgenius.com/go/10C054CAEAE28AAF8C61-55865852-final>
2. Had your topic proposal (see below) already approved by your TA. This means that you should have contacted your TA around a week before that already, in case your initial proposal isn't approved and you need more time.

If you do not meet the April 8 deadline, marks will be deducted from your presentation evaluation. In particular, no presentations will be allowed with topics that have not been approved well in advance (hence, not getting your topic approved in time means getting 0% on your presentation evaluation).

Presentation topic, structure and evaluation

The main goal of the presentation is to present an interesting problem and its algorithmic solution. The solution should rely on the algorithms and techniques learned in class, and specifically, we expect it to rely either on FFT, or on max-flow, or on linear programming. Solutions using other algorithms and techniques may be considered as exceptions, but only if there is an extremely compelling reason (e.g., there is an ingenious idea making the solution very interesting).

You are encouraged to look up interesting problems online and in textbooks, and to choose a topic creatively and according to your personal interests. We will be liberal in allowing unconventional settings, such as approximation problems, online settings, problems where space/memory complexity is the bottleneck, and more.

Structure and evaluation. You must use slides, in PDF format. The presentation's structure is at your discretion, but we strongly recommend to consider (as a default) the following blueprint:

1. The problem: Presenting intuition and motivation for the problem, then its formal definition.
2. Result: Formal statement for the existence of an efficient algorithm solving the problem (including correctness guarantee and runtime complexity).
3. Algorithm: High-level intuition for the main idea, and then a full proof that includes a description of the algorithm and its analysis.

The evaluation criteria include (but aren't limited to) completeness, clarity and correctness of your definitions and result statements, of the presentation of intuition and ideas, and of the algorithm's description and analysis.

The presentations should last 10–15 minutes, and you will be *strictly limited to 15 minutes* (hard deadline). Presentations that are not over at the 15-minute mark will be stopped, and will lose marks in the evaluation. (This is a basic skill in giving presentations: Respecting the time constraints of your audience!)

Topic proposal guidelines

Aim to email your TA by April 1st with a topic proposal. The email must come from both students (i.e., the one not sending the email must be CC'ed), and the topic proposal should include:

1. The names and email addresses of both students.
2. A description of the problem: The problem's name and a (short) description.
3. A short description of the algorithm you intend to use for solving the problem. If this algorithm doesn't use FFT, max-flow, or linear programming, please provide an extremely compelling reason for why this is a suitable topic for presentation.
4. The specific materials (texts, videos, presentations) you will rely on. This needs to be 100% explicit, including links (or file attachments), page numbers, and references to specific result statements.

As mentioned above, if you send a proposal at the last minute before April 8, and the proposal isn't accepted, you will lose marks for late submission. We do not guarantee a response in time unless you send the proposal by April 1st.

First-come-first-served. Only one pair of students will be allowed to present any given topic. Whichever pair of students gets their proposal accepted first will be allowed to present the topic, and other pairs won't.